2) Phase Modulators:

Upper Branch $+\Delta\varphi$ phase shift

Lower Branch $-\Delta\varphi$ phase shift

$$E_1' = E_1 \cdot e^{i\Delta\varphi} = E_{in}\cdot\sqrt{\frac{p}{p+1}}\cdot e^{i\Delta\varphi}$$

$$E_2' = E_2 \cdot e^{-i\Delta\varphi} = E_{in}\sqrt{\frac{1}{1+p}}\; e^{-i\Delta\varphi}$$

(3) Power Combiner

$$\frac{E_1}{\sqrt{2}} + \frac{E_2}{\sqrt{2}} = E_{out} \quad \rightarrow \quad \frac{1}{\sqrt{2}} \text{ due to Energy conservation}$$

$$E_{out} = \frac{E_{in}}{\sqrt{2}}\cdot\left(\sqrt{\frac{p}{p+1}}\cdot e^{i\Delta\varphi} + \sqrt{\frac{1}{p+1}}\; e^{-i\Delta\varphi}\right)$$

$$\frac{E_{out}}{E_{in}} = \frac{1}{\sqrt{2}}\;\frac{1}{\sqrt{p+1}}\left(\sqrt{p}\cdot e^{i\Delta\varphi} + e^{-i\Delta\varphi}\right)$$

$$T = \left(\frac{E_{out}}{E_{in}}\right)^2 = \frac{P_{out}}{P_{in}} = \left(\frac{E_{out}}{E_{in}}\right)\cdot\left(\frac{E_{out}}{E_{in}}\right)^{*}$$

* ← complex conjugate

$$= \frac{1}{2}\cdot\frac{1}{p+1}\cdot\left(\sqrt{p}\,e^{+i\Delta\varphi} + e^{-i\Delta\varphi}\right)\left(\sqrt{p}\cdot e^{-i\Delta\varphi} + e^{i\Delta\varphi}\right)$$

$$= \frac{1}{2}\;\frac{1}{p+1}\left(p+1+\sqrt{p}\,e^{2i\Delta\varphi} + \sqrt{p}\,e^{-2i\Delta\varphi}\right)$$

$$T = \frac{1}{2}\left(1 + \frac{2\sqrt{p}}{p+1}\cdot\cos 2\Delta\varphi\right)$$

$$T_{max} = \cos 2\Delta\varphi \overset{!}{=} 1 \quad \rightarrow \quad T_{max} = \frac{1}{2}\left(1+\frac{2\sqrt{p}}{p+1}\right)$$

$$T_{min} = \cos 2\Delta\varphi \overset{!}{=} -1 \quad \rightarrow \quad T_{min} = \frac{1}{2}\left(1-\frac{2\sqrt{p}}{p+1}\right)$$

$T_{max} + T_{min} = 1$ & lossless device!

b) U_π is the voltage needed to introduce an phaseshift of π!

It can be determined by finding the Minimum in the power -> voltage Diagramm of the Modulator.

From Diagramm: $U_{max} = 2V$ $U_{min} = 6.5V$ $U_\pi = 6.5V - 2V = 4,5V$

find L:

$$U_\pi = \frac{\pi \lambda x}{4 x n_e^3 \cdot L \cdot r_{33}} \rightarrow L = \frac{\pi \cdot \lambda \cdot x}{4 \cdot n_e^3 \cdot U_\pi \cdot r_{33}} \qquad \frac{m}{V \cdot \frac{m}{V}}$$

$n_{eff} = 2,15$ $r_{33} = 30.9 \frac{pm}{V}$ -> for $LiNbO_3$

$$L = \frac{\pi \cdot 1.55\,\mu m \cdot 10\,\mu m}{4 \cdot 2,15^3 \cdot 4,5V \cdot 30,9 \frac{pm}{V}} = 8,81\,mm$$

c) T_{min} can't be found from the linear scaled diagramm.

↳ log scale:

$$T_{min,dB} = P_{min,dB} - P_{in,dB} = -37dBm - 10 \cdot \log\left(\frac{200\,\mu W}{1mW}\right) = -37dBm - (-7dBm) = -30dB$$

$$\rightarrow T_{min} = 1 \cdot 10^{-3}$$

$$T_{max,dB} = 10 \cdot \log\left(\frac{P_{max}}{1mW}\right) - P_{in,dBm} = 10 \cdot \log\left(\frac{195\,\mu W}{1mW}\right) - (-7dBm) = -7,1dBm + 7dBm = -0,1dB$$

$$\rightarrow T_{max} = 0.977$$

$$ER = T_{max} - T_{min} = -0,1dB + 30dB = 29,9\,dB$$

ER ≙ extinction-ratio

d) Ideal case: $T_{max} + T_{min} = 1$

Here: $T_{max} + T_{min} = 0.977 + 1 \cdot 10^{-3} = 0,978 \neq 1$ -> losses in wave guides!

$T_{max} + T_{min} = \alpha \left(T_{max}' + T_{min}'\right) = \alpha = 0,978$

$$T_{max}' = \frac{T_{max}}{\alpha} = \frac{0.977}{0.978} = 0.999 \quad , \quad T_{min}' = \frac{T_{min}}{\alpha} = 0.001$$

$$T_{max}' = \frac{1}{2}\left(1 + \frac{2\sqrt{p}}{p+1}\right) \quad , \quad T_{min}' = 1/2\left(1 - \frac{2\sqrt{p}}{1+p}\right)$$

$$T_{max}' T_{min}' = \frac{1}{4}\left(1 - \frac{4p}{(p+1)^2}\right)$$

$$4\, T_{max}' T_{min}' (p+1)^2 = (p+1)^2 - 4p = p^2 + 2p + 1 - 4p = p^2 - 2p + 1 = (p-1)^2$$

$$\Rightarrow \quad 2 \cdot \sqrt{\left(T_{max}' T_{min}'\right)} \cdot (1+p) = (1-p)$$

$$p = \frac{1 - 2 \cdot \sqrt{T_{max}' T_{min}'}}{1 + 2 \cdot \sqrt{T_{max}' T_{min}'}} = \frac{1 - 2\sqrt{0.999 \cdot 0.001}}{1 + 2\sqrt{0.999 \cdot 0.001}} = 0.88$$

$$\Rightarrow \frac{P_2}{P_{in}} = \frac{1}{1+p} = \frac{1}{1 + 0.88} = 0.53$$

$$\Rightarrow \frac{P_R}{P_{in}} = 1 - \frac{P_2}{P_{in}} = 0.47$$

Problem 15

Indium $\quad x = 0.53$

$E_{g,\,InGaAs} = 0.75\ eV \qquad n_{InGaAs} = 3.56$

$E_{g,\,InP} = 1.35\ eV \qquad n_{InP} = 3.4$

$$E_{Photon} = h \cdot f = h \cdot \frac{c}{\lambda}$$

$$1\,eV = 1.602 \cdot 10^{-19}\,J$$

Charge per electron

$1.35\,eV \cdot 1.602 \cdot 10^{-19} = 2.1627 \cdot 10^{-19}\,J$

$0.75\,eV \cdot \text{—''—} = 1.2015 \cdot 10^{-19}\,J$

} Energy bounds!

$$E = hf = h\,\frac{c}{\lambda} \qquad \lambda = \frac{hc}{E}$$

$2.1627 \cdot 10^{-19}\ J$
$1.2015 \cdot 10^{-19}\ J$
} Energy bounds! $\lambda = \frac{hc}{E}$

$$\lambda_{lower} = \frac{6.626 \cdot 10^{-34}\ Js \cdot 3e8\ \frac{m}{s}}{1.2015 \cdot 10^{-19}\ J} = 1.65\ \mu m$$

$$\lambda_{upper} = \frac{6.626 \cdot 10^{-34}\ Js \quad 3e8\ \frac{m}{s}}{2.1627 \cdot 10^{-19}\ J} = 919,12\ nm$$

P_4

b)

$\lambda_0 = 1300\ nm$

$P_0 = 10\ \mu W$

P_3 $3.56 = n_{InGaAs}$

P_2

P_1 $3.4 = n_{InP}$

$1.00 = n_{air}$

P_0

$$P_3 = P_0 \left(1 - \left(\frac{n_{air} - n_{InP}}{n_{air} + n_{InP}}\right)^2\right) \cdot e^{-\alpha \cdot z} \cdot \left(1 - \left(\frac{n_{InP} - n_{InGaAs}}{n_{InP} + n_{InGaAs}}\right)^2\right)$$

$$= 10\mu W \left(1 - \left(\frac{1.00 - 3,4}{1.00 + 3.4}\right)^2\right) \cdot e^{-0.25 \cdot 10^2\ m^{-1} \cdot 500 \cdot 10^{-6}\ m} \cdot \left(1 - \left(\frac{3,4 - 3.56}{3.4 + 3,56}\right)^2\right) = 6.93\ \mu W$$

$\left(1 - \left(\frac{3,4 - 3.56}{3.4 + 3,56}\right)^2\right) \approx 0$

→ most reflection happening at the air - InP interface.

→ wth? anti reflection coating!

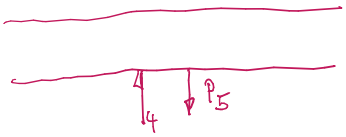
c)

$$P_{abs} = P_{InGaAs} \cdot \left(1 - e^{-\alpha \cdot z}\right)$$

$$= 6,93\ \mu W \cdot \left(1 - e^{-5 \cdot 10^6\ m^{-1} \cdot 1 \cdot 10^{-6}\ m}\right) = 6,88\ \mu W\ !$$

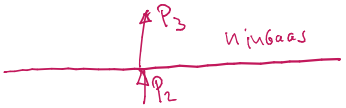
6.93 µW $\quad P_4 = 6{,}93\,\mu W \cdot e^{-5\cdot 10^6 m^{-1} \cdot 1\cdot 10^{-6} m} = 4{,}67\,nW$

$P_5 = 4{,}67\,nW \quad 0.98 = 4{,}58\,nW$

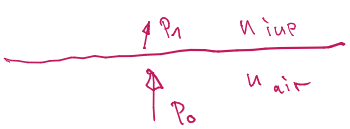


$P_{abs_2} = 4{,}58\,nW \cdot \left(1 - e^{-5\cdot 10^6 m^{-1} \cdot 1\cdot 10^{-6} m}\right)$

$= 4{,}55\,nW$



$\eta_1 = \frac{6{,}93\,\mu W}{10\,\mu W} = 69{,}3\,\%$



$\eta_2 = \frac{6.93\,\mu W + 4{,}58\,nW}{10\,\mu W} = 69{,}35\,\%$

Photo current ?

remember $\quad E = h \cdot f$

$\eta_p = \frac{P}{h \cdot f}$ → number of photons per second

now, quantum efficiency η, fraction of incident photons that successfully eject photoelectrons:

$$\eta \cdot n = \frac{\eta \cdot P}{hf}$$

→ current as charge per unit time; with every electron carrying charge e:

$$I = \frac{\eta \cdot P e}{h \cdot f} = \boxed{\frac{\eta \cdot P e \lambda}{hc}}$$

69,35 %

$$\frac{\eta \cdot P_e \cdot \lambda}{h c} = I \qquad I_2 = \frac{0.6935 \cdot 10\,\mu W \cdot 1.602 \cdot 10^{-19} C \cdot 1300\,nm}{6.626 \cdot 10^{-34} Js \cdot 3e8 \frac{m}{s}} \qquad \frac{C}{s} = A \;\checkmark$$

$$I_2 = 7{,}27\ \mu A$$

e) $\tau_{RC} = (R_p + R) \cdot C_D$, with $(R_p + R) = 50\,\Omega$

$$C_p = \varepsilon_0 \cdot \varepsilon_r \cdot \frac{A}{z} = \varepsilon_0 \cdot n^2_{InGaAs} \cdot \frac{\pi \frac{D^2}{4}}{z} = 8.8541\,9 \frac{pF}{m} \cdot (3{,}56)^2 \cdot \frac{\pi \cdot \frac{20\mu m^2}{4}}{1\mu m}$$

(z = height)

$$= 35{,}25\ fF$$

$$\tau_{RC} = 50\,\Omega \cdot 35{,}25\ fF = 1.7625\ ps$$

f) $v_n = 1 \cdot 10^5 \frac{m}{s}$ $\quad t_n = \frac{z}{v_n} = \frac{1\mu m}{1 \cdot 10^5 \frac{m}{s}} = 10\,ps \rightarrow$ deals with autc!

$$\tau_n = \frac{t_n}{0.88\pi} = 3.617\,ps$$

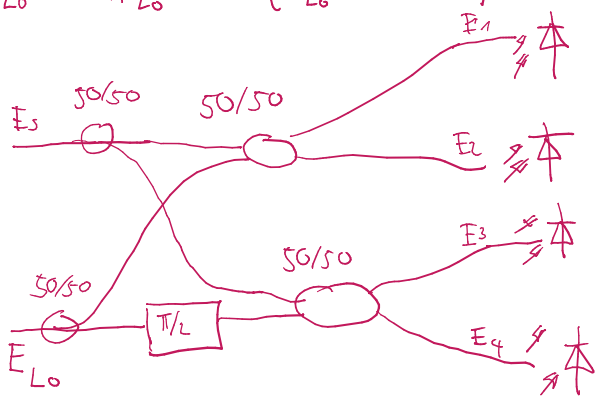
$$f_{3dB} = \frac{1}{2\pi} \cdot \omega_{3dB} = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{\tau_n^2 + \tau_{Rc}^2}} = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{(3{,}617_{ps}^2 + 1.765_{ps}^2)}} =$$

39,56 GHz

Problem 16

$E_S = A_S \cdot \cos(\omega_S t + \phi_S(t)) \rightarrow$ signal field

$E_{LO} = A_{LO} \cdot \cos(\omega_{LO} t + \phi_{LO}(t)) \rightarrow$ local oscilator field.



where is the necessity for coherent detection?

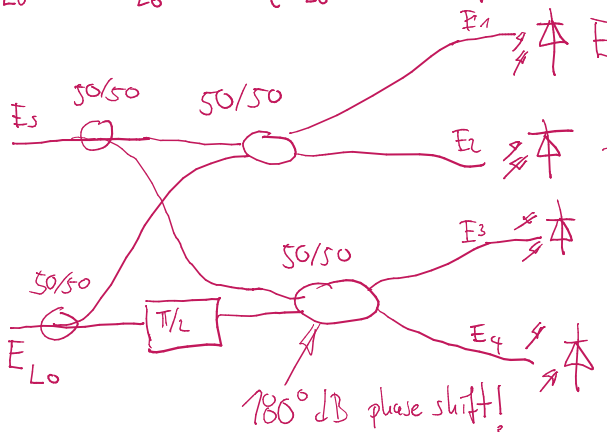
↳ Higher Dimension for transmitting data, intensity & phase dimension!

Phase can only be detected if one has a reference optical signal, a relative phase between those two oscillating fields can be measured!

$E_S = A_S \cdot \cos\left(\omega_S t + \phi_S(t)\right)$

$E_{LO} = A_{LO} \cdot \cos\left(\omega_{LO} t + \phi_{LO}(t)\right)$

condition: all fibers are length matched, and no other phase delays are introduced (refr. index differences ... usw)



$E_1 = 1/2 \left(A_S \cdot \cos\left(\omega_S t + \phi_S(t)\right) + A_{LO} \cdot \cos\left(\omega_{LO} t + \phi_{LO}(t)\right) \right)$

$E_2 = 1/2 \left(A_S \cdot \cos\left(\omega_S t + \phi_S(t)\right) - A_{LO} \cdot \cos\left(\omega_{LO} t + \phi_{LO}(t)\right) \right)$

$E_3 = 1/2 \left(A_S \cdot \cos\left(\omega_S t + \phi_S(t)\right) + A_{LO} \cdot \cos\left(\omega_{LO} t + \pi/2 + \phi_{LO}(t)\right) \right)$

$E_4 = 1/2 \left(A_S \cdot \cos\left(\omega_S t + \phi_S(t)\right) - A_{LO} \cdot \cos\left(\omega_{LO} t + \pi/2 + \phi_{LO}(t)\right) \right)$

→ fields Again, cant be measured $I \sim |E|^2$

$$I_1 = \frac{A_S^2}{4}\left\{1 + \cos\left(2\omega_S t + 2\phi_S(t)\right)\right\} + \frac{A_{LO}^2}{4}\left\{1 + \cos\left(2\omega_S t + 2\phi_S(t)\right)\right\} +$$

0 → Lowpass of photodiodes

0 Lowpass of photodiodes

$$\frac{A_S A_{LO}}{2}\left\{\cos\left[(\omega_S - \omega_{LO})t + \phi_S(t) - \phi_{LO}(t)\right] + \cos\left[(\omega_S - \omega_{LO})t + \phi_S(t) + \phi_{LO}(t)\right]\right\}$$

0 Lowpass from photodiode

Low pass filtering : ω_S & ω_{LO} usually oscillate at 192 THz,

Photodiode 3dB bandwidth is max max! @ 400 GHz

→ so cosine terms of these elements = 0

$DC_{signal} = \frac{A_S^2}{4}$ → background offset from Signal

$DC_{\text{local oscillator}} = \frac{A_{LO}^2}{4}$ → background offset from LO

$\text{Beat Signal} = \frac{A_S A_{LO}}{2} \cos\left(\omega_{IF} t + \Delta\phi(t)\right)$

$\omega_{IF} = \omega_S - \omega_{LO}$ → intermediate frequency, if ≈ 0 → homodyne detection

$\Delta\phi(t) = \phi_S(t) - \phi_{LO}(t)$ (phase difference)

$$I = \frac{A_S^2}{4} + \frac{A_{LO}^2}{4} + \frac{A_S A_{LO}}{2} \cdot \cos\left(\omega_{IF} t + \Delta\phi(t)\right) = I_S + I_{LO} + 2 \cdot \sqrt{I_S I_{LO}} \cdot$$

$$I = I_S + I_{LO} + 2 \cdot \sqrt{I_S I_{LO}} \cdot \cos\left(\omega_{IF} + \Delta\phi\right)$$

beat term is carried from a strong local oscillato and thus gives an advantage of direct detection.